Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/17/2006
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Forced Flow Ignition and Flame Spread Test (FIST)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/20/2004	**End Date:**	07/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA02G		
Performance Goal No.:			
Performance Goal Text:			
	The Forced flow Ignition and Flame Spread Test (FIST) aims to study the effects of gravity on the flammability diagrams of solid combustible materials, and the fire properties derived from them. The final objective of the research is to develop a new flammability apparatus that would better reflect the potential ambient conditions of space-based environments, i.e., microgravity, low velocity flow, oxygen concentrations below and above that of air, and use it to obtain more appropriate flammability diagrams for practical materials (plastics, cable jackets, electronic circuit boards, etc.) that will be found in space-based facilities. The strong possibility of an accidental fire occurring in a space-based facility during a long mission is a primary safety concern. Consequently, it is of critical importance to characterize the fire properties of materials that are used. FIST is similar to the ASTM E-1321 LIFT apparatus, except the fuel sample is mounted in small-scale wind tunnel so that a low-velocity oxidizer flow of specified oxygen concentration can be forced over the sample surface during the test.		

Task Description:	The FIST protocol includes both ignition and flame spread testing. For an ignition test, a 30 mm by 30 mm test specimen is embedded flush in the substrate of the wind tunnel so that the gas flow boundary layer generated over its surface is not disrupted. Two radiant heaters are used to irradiate the sample with an approximately uniform heat flux of known intensity. Ignition is induced at fuel sample's trailing edge with an electrically heated wire coil that acts as a pilot. Measured quantities for a particular material include the time to ignition as a function of ambient oxygen concentration and flow velocity, as well as the critical (minimum) mass and heat flux for ignition. Alternatively, flame spread tests can be conducted using larger samples (100 mm long by 30 mm wide). The sample is exposed to a heat flux that decays in the longitudinal direction. The sample is ignited locally at one end, and a flame spreads over teh sample surface. The rate of flame spread is measured by digitizing infrared images and taking the slope of the flame position vs. time. The ignition and flame spread data are plotted on a flammability diagram which is a measure of a material's overall flammability. To complement the experimental program, theoretical modeling of the ignition, burning, and flame spread process is performed. This modeling includes a description of the solid-phase heating and fuel generation process through a customized pyrolysis model. The pyrolysis model is coupled to the CFD-based Fire Dynamics Simulator for calculation of gas-phase phenomena (fluid mechanics, combustion, heat transfer, radiation, etc.). The combined model is used to calculate ignition, burning, and flame spread processes for different materials, environmental conditions, and geometries. One of the principal purposes of the modeling effort is to extend via modeling the experimental FIST results to other conditions that have not been tested, i.e. different O2, velocity, or gravity levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Composites are finding increased use in the transportation and construction industries. One barrier to their widespread adoption is the increased flammability of polymer based composites when compared with other materials. In the process of developing a test methodolgy for flammability of composites intended for use in space structures, the FIST project is developing a better understanding of composite materials' flammability useful to ground based design and manufacture.
Task Progress:	During FY2006, studies were conducted to examine the variation of the critical mass loss rate at ignition with freestream oxygen content, flow velocity, and applied radiative heat flux. The critical mass loss rate at ignition is primary component of a material's overall ignitability and has important implications for microgravity fire safety. Experimental data were compared to a simplified theoretical model with good agreement between the predicted and measured values. This model was described and its predictions were presented in the FY2005 report. Although the theoretical model successfully predicted the variation of the critical mass loss rate at ignition with freestream oxygen content and velocity, it predicted no variation of the critical mass loss rate at ignition with heat flux. However, experimental FIST measurements clearly showed an increase in the critical mass loss rate at ignition with heat flux, a trend also observed by other researchers. Since the model predicted no variation in the critical mass flux at the fire point with heat flux level, but the experimental FIST results showed an increase in the critical mass flux at the fire point with the applied heat flux level, additional experiments were conducted to isolate the influence of temperature and oxygen as factors in fuel production. PMMA samples 3 mm and 10 mm thick were used to study the effect of temperature. The role of oxygen was studied by modifying the surface of the 10 mm thick PMMA samples with 1 mm diameter (0.5 mm deep) dimples spaced at approximately 5 dimples per sq cm. At low heat flux levels, modifying the surface results in only a mild reduction in the critical mass loss rate at ignition. For higher heat flux levels, these modifications result in a significant lowering of the critical mass loss rate at ignition. This is explained on the basis of oxygen participation in the polymer decomposition process, which is known to produce radical species. Specifically, the enhanced formation of these radical species attributed to increased oxygen access may enhance ignition. Work related to flame spread involved analysis of the microgravity flame-spread experiments conducted on the KC-135 aircraft in October 2004. These results demonstrate that microgravity flame spread rates may be faster than those under normal gravity conditions when exposed to an external heat flux at oxygen concentrations of 24%. These elevated flame spread rates result from a decrease in oxidizer velocity below that induced by gravity which increases residence time, thereby minimizing kinetic effects associated with short chemical times. This allows the flame to propagate in a region where the flame spread mechanism is dominated by heat transfer effects. As a complement to the flame spread experiments, simplified analytical and theoretical models have been developed. A simple analytical theory based on transition from a heat-transfer controlled regime and a chemical kinetics controlled regime has been used to predict normal gravity flame spread rates as well as microgravity KC-135 experiments with excellent agreement between the predictions and the experimental data. A numerical flame spread model is currently being developed. The model involves using the CFD-based Fire Dynamics Simulator (FDS) code developed by the National Institute of Standards and Technology (NIST) coupled to a pyrolysis model developed in our laboratory to simulate the effects of oxygen concentration and freestream velocity on opposed flow flame spread rates over thermally thick samples of PMMA. The FDS thermoplastic pyrolysis model was replaced with a model that considers subsurface pyrolysate generation by incorporating a distributed pyrolysis reaction. Additional modifications were made to reduce the over-predicted flame radiation (which resulted in unreasonably low flame temperatures) and to reduce CPU time by adopting a quasi-steady formulation. Calculations were conducted under forced flow conditions only, i.e. in the absence of gravity. Good agreement between the model predictions and the experimental data was obtained at an oxygen concentration used for model "calibration", but the flame spread rate is underpredicted at higher oxygen mass fractions and underpredicted at lower oxygen mass fractions. Future work during FY2007 will involve modeling oxygen concentration effects on gas and solid-phase kinetics to improve model predictions. The numerical flame spread modeling described above used a generalized pyrolysis model that has been developed over several years. During FY2006, work continued on development of this model, intended for simulating the gasification of noncharring, charring, and intumescent materials, as well as smoldering in porous media. Conservation equations are solved for gaseous and condensed phase mass and species, solid phase energy, and gas-phase momentum. An arbitrary number of gas-phase and condensed-phase species can be accommodated, each having its own temperature-dependent thermophysical properties. The user may specify any number of reactions. Both in-depth radiation transfer through a semi-transparent medium as well as radiation transport across pores are considered. Melting is modeled using an apparent specific heat. All volatiles generated inside the solid escape to the ambient with no resistance to flow unless the pressure solver is invoked to solve for the pressure distribution in the solid, with the resultant flow of volatiles

	calculated according to Darcy's law. The user may invoke a gas-phase convective-diffusive solver that determines the composition of the volatiles, including diffusion of species from the ambient into the solid. Thus, the actual composition of the pyrolysate vapors escaping from the solid can be predicted. To aid in determination of the required material properties, the model has been coupled to a genetic algorithm optimization procedure that can be used to estimate material properties from laboratory tests. The genetic algorithm has been parallelized and run simultaneously on as many as 50 networked PCs to reduce CPU time. As mentioned earlier, the model has been integrated with NIST Fire Dynamics Simulator v4.06 so that it may be applied as a boundary condition in 2D and 3D simulations of ignition, flame spread, and fire growth. Although the pyrolysis model is one-dimensional, when it is applied as a boundary condition in FDS, two-dimensional heat conduction effects can be calculated. The standalone pyrolysis model has been used to simulate the gasification of a noncharring solid, a charring solid, an intumescent coating, and smoldering combustion in a porous medium, with generally good results. The flexibility of this model to simulate different materials is important for space-based applications where a variety of materials are encountered. A paper describing this work is currently being prepared.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Books/Book Chapters	Lautenberger C, Torero J, Fernandez-Pello C. "Understanding materials flammability." in "Flammability testing of materials used in construction, transport, and mining." Ed. V. Apte. Cambridge, UK : Woodhead Publishing Ltd., p. 1-21, 2006., Jan-2006
Books/Book Chapters	Lautenberger C, Fernandez-Pello C. "Pyrolysis Modeling, Thermal Decomposition, and Transport Processes in Combustible Solids." in "Transport Phenomena in Fires." Ed. M. Faghri and B. Sunden. Billerica, MA : WIT Press/Computational Mechanics , in press, 2007., Jan-2007
Dissertations and Theses	Hernandez JA. "Effect of Sample Thickness on Mass Loss Rate of Combustible Solids at the Flashpoint and Firepoint." Masters thesis, UC Berkeley, May 2006. , May-2006
Papers from Meeting Proceedings	Lautenberger, C.; McAllister, S.; Rich, D.; Fernandez-Pello, C. "Modeling the Effect of Environmental Variables on Opposed-flow Flame Spread Rates with FDS." International Congress on Fire Safety in Tall Buildings, University of Cantabria, Santander, Spain, October 2006. University of Cantabria, Santander, Spain, 2006. , Oct-2006
Papers from Meeting Proceedings	Rich, D., Lautenberger, C., Torero, J.L., Quintiere, J.G., Fernandez-Pello, A.C. "Mass Loss Rate of Combustible Solids at Piloted Ignition." 31st International Symposium on Combustion, Heidelberg, August 2006. 31st International Symposium on Combustion, Heidelberg, August 2006. , Aug-2006
Papers from Meeting Proceedings	Rich, D., Lautenberger, C., McAllister, S., Fernandez-Pello, A.C. "Flame Spread Rates Over Samples of Polymer and Polymer Glass Composites." Western States Section/Combustion Institute Spring 2006 Meeting, Boise State University, Boise, Idaho, March 2006. Boise, Idaho : Boise State University, 2006. , Mar-2006